



Reduction bound and hop purpose

$$\text{Adv}_{\text{hide}}(\mathcal{A}) \leq \text{Adv}_{\text{PH-MA-ABE}}(\mathcal{B}_1) + \text{Adv}_{\text{COM}}(\mathcal{B}_2) + \text{Adv}_{\text{PAD}}(\mathcal{B}_3) + \text{Adv}_{\text{SIG}}(\mathcal{B}_4)$$

$$+ \text{Adv}_{\text{CRED}}(\mathcal{B}_{5a}) + \text{Adv}_{\text{ZK}}(\mathcal{B}_{5b}) + \epsilon_R + \text{negl}(\lambda)$$

B₁: policy/attribute hiding

B₂: commitment consistency

B₃: request-type padding

B₄: audit-chain unforgeability

B_{5a}: key-credential forgery

B_{5b}: proof soundness